Delivering on the Promise

BazaarLink delivery performance and customer satisfaction

Capstone Project: E-Commerce Delivery & Customer Experience Analytics
Presented by: Nepstina S, Business / Data Analyst

Data: Olist Brazilian e-commerce dataset (99,441 orders, 2016–2018) · Tools: Excel · MySQL · Python · Power BI · Tableau · Jira · Confluence

PERFORMANCE ANALYSIS · 2016–2018

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The Problem

Growth Has Stalled

Bad delivery leads to bad reviews — and customers don't return. The feedback loop is eroding the business.

No Single Source of Truth

Each team had its own numbers. Nobody agreed on the on-time rate, making it impossible to act with confidence.

The Central Question

What is late delivery costing us — and where do we act first?

Why This Matters

Delivery experience is the single most controllable driver of customer satisfaction in e-commerce. Every late parcel is a review risk, a churn event, and a lost repeat purchase.

This analysis puts a number on that cost — and points to exactly where to intervene.

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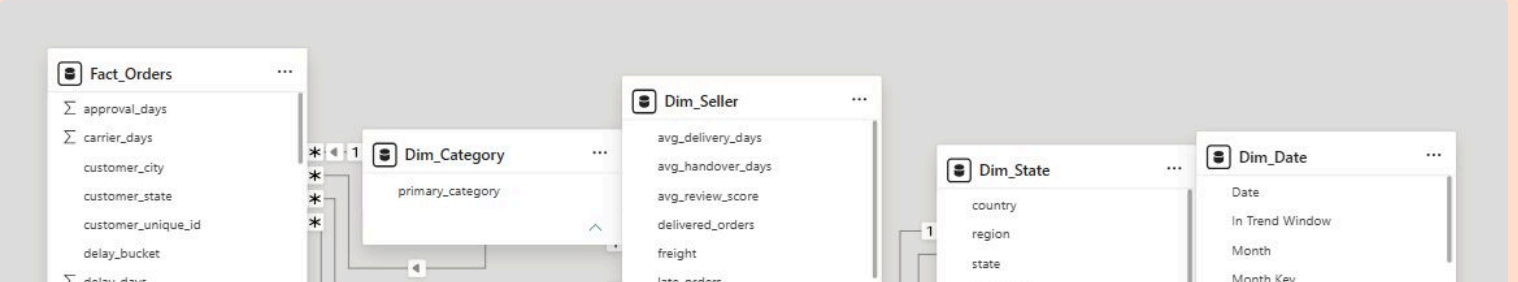
Data, Approach & Headline Numbers

Methodology

99,441 orders across 9 linked tables, covering 2016–2018 (Olist dataset).

Cleaned in **Python** (48 data issues logged) → **MySQL** (25 queries, 3 views) → Python statistics → **Power BI, Tableau, Excel**.

One KPI definition set: all tools match exactly.



93.2%

On-Time Rate

12.6

Avg Delivery Days

4.09

Avg Review Score

14.7%

Negative Reviews

3.1%

Repeat Customers

R$13.2M

Total Revenue

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Late Delivery Is Strongly Linked to Bad Reviews

Statistical analysis across 99,441 orders confirms the relationship is not coincidental — it is significant and severe.

Star Rating Gap

Late orders average **2.27 stars** vs **4.29 stars** for on-time orders.

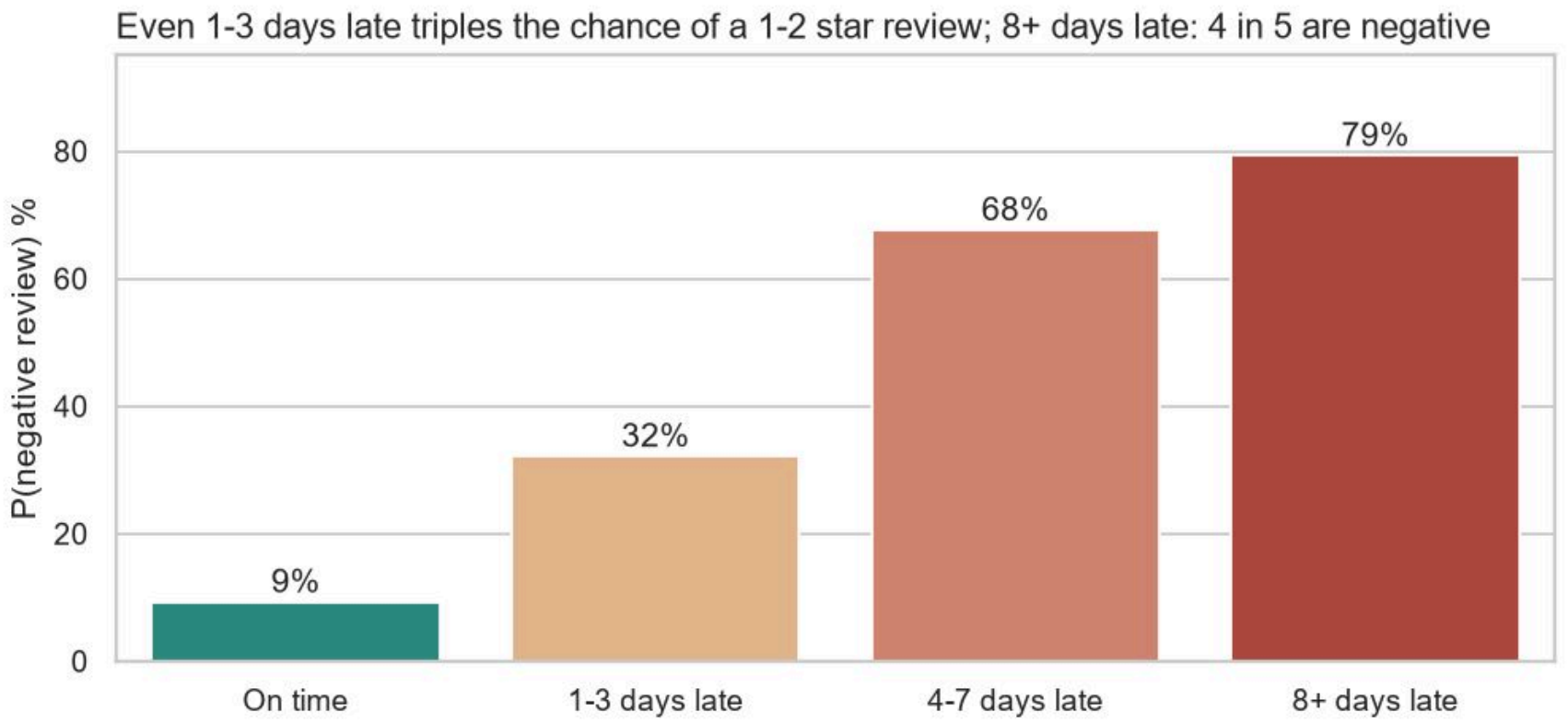
Mann-Whitney p < 0.001

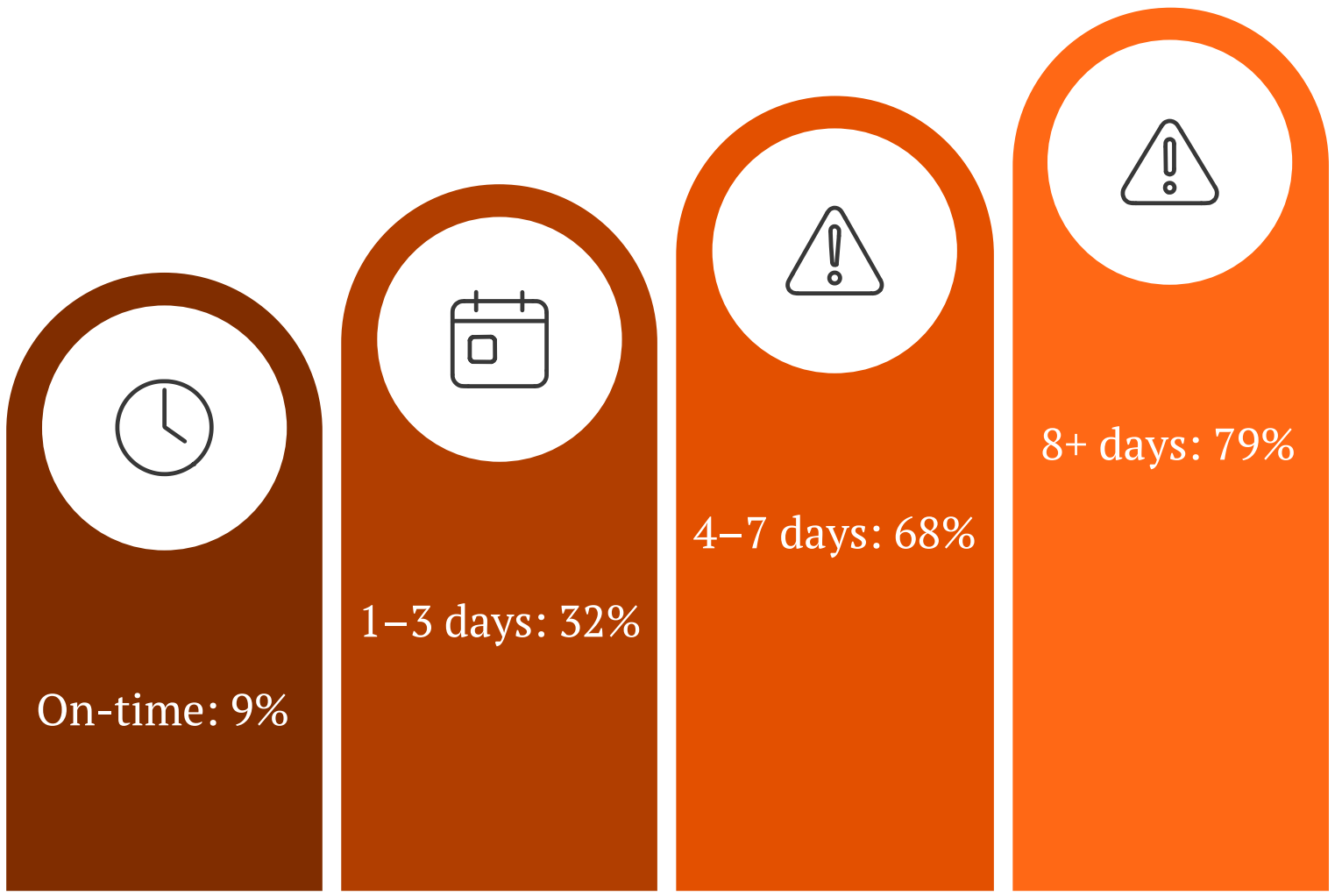
6.7× More Likely

A late order is **6.7×** as likely to receive a 1–2 star review compared to an on-time order.

Escalating Risk

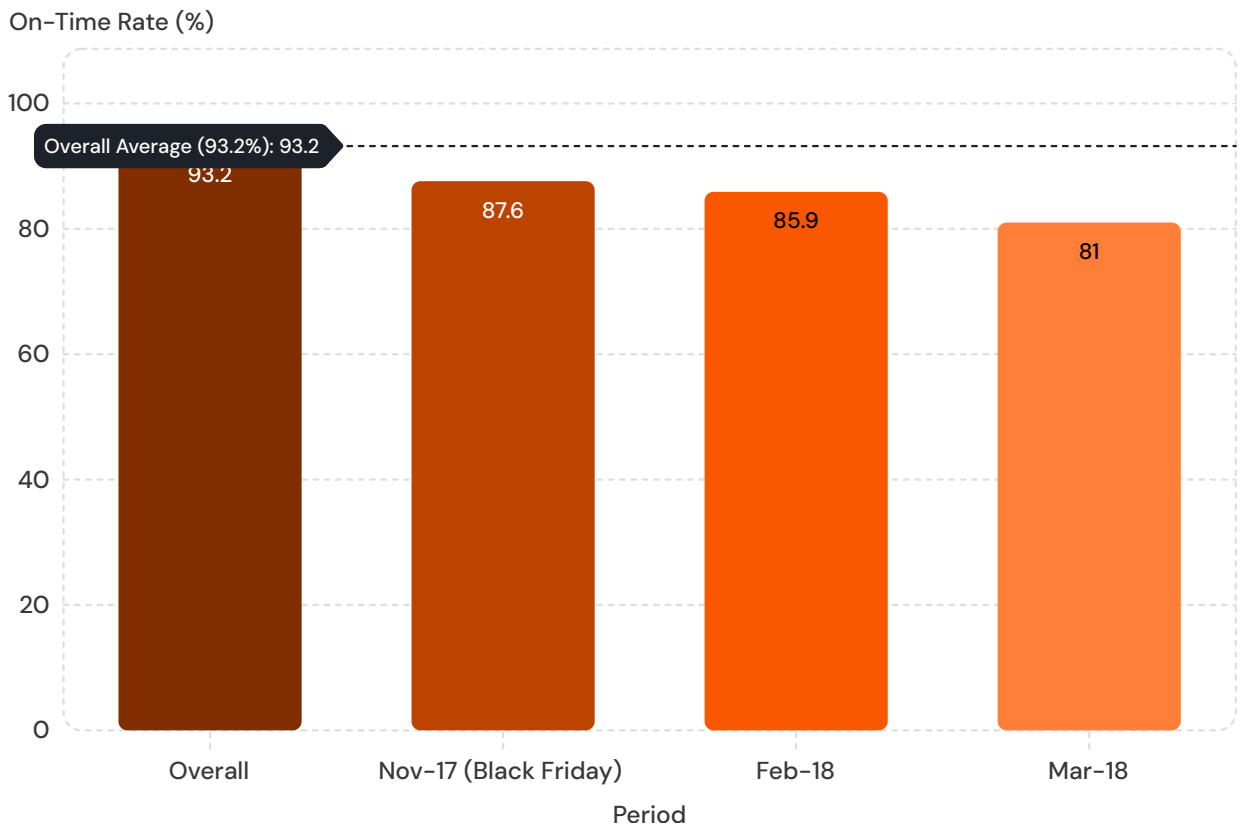
9% negative on time → **32%** if 1–3 days late → **68%** → **79%** if 8+ days late.





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When: On-Time Collapses at Peaks

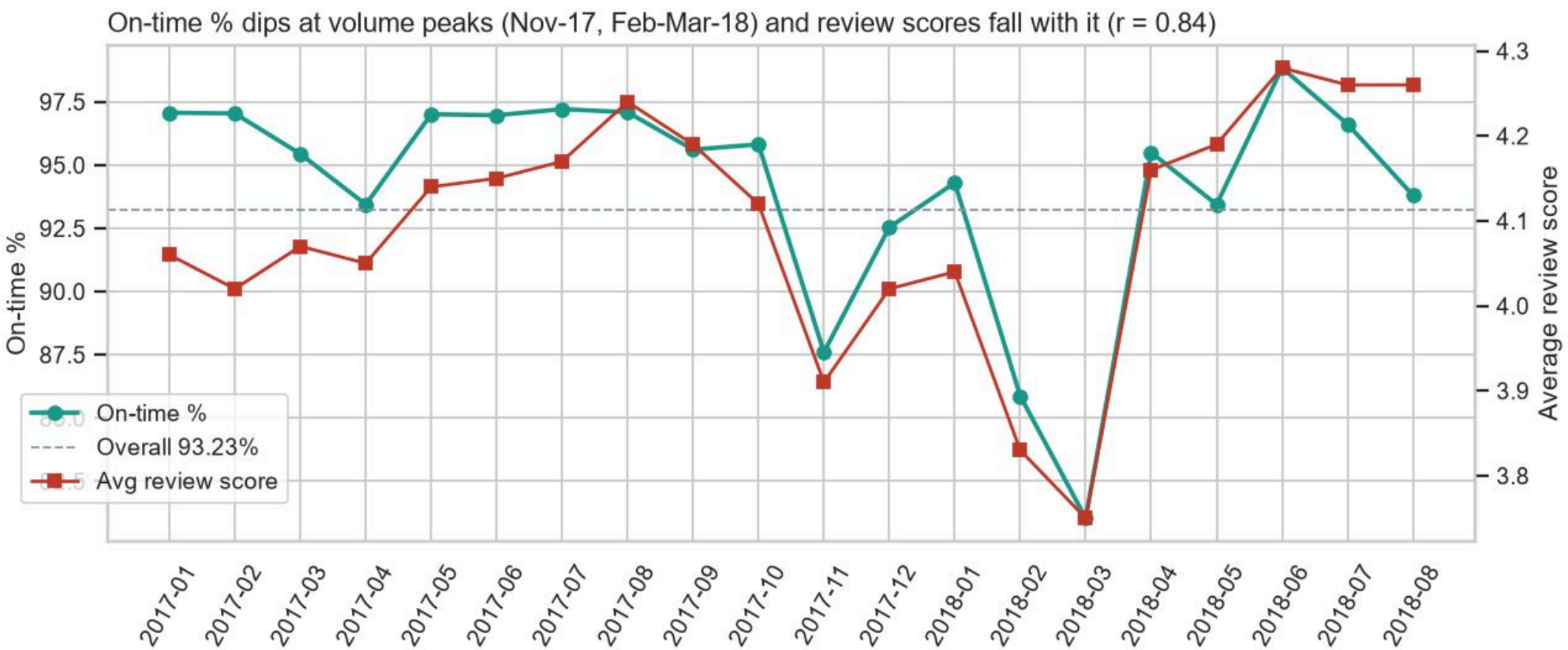


The Pattern Is Clear

The overall on-time rate of **93.2%** masks dangerous dips during demand peaks.

- **87.6%** in November 2017 — Black Friday surge coincided with the drop
- **85.9%** in February 2018 —high-volume month
- **81.0%** in March 2018 — the worst month on record

Peak periods are when customers need reliability most — and when the system fails them.



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Where Time Is Lost



The Carrier Leg Dominates

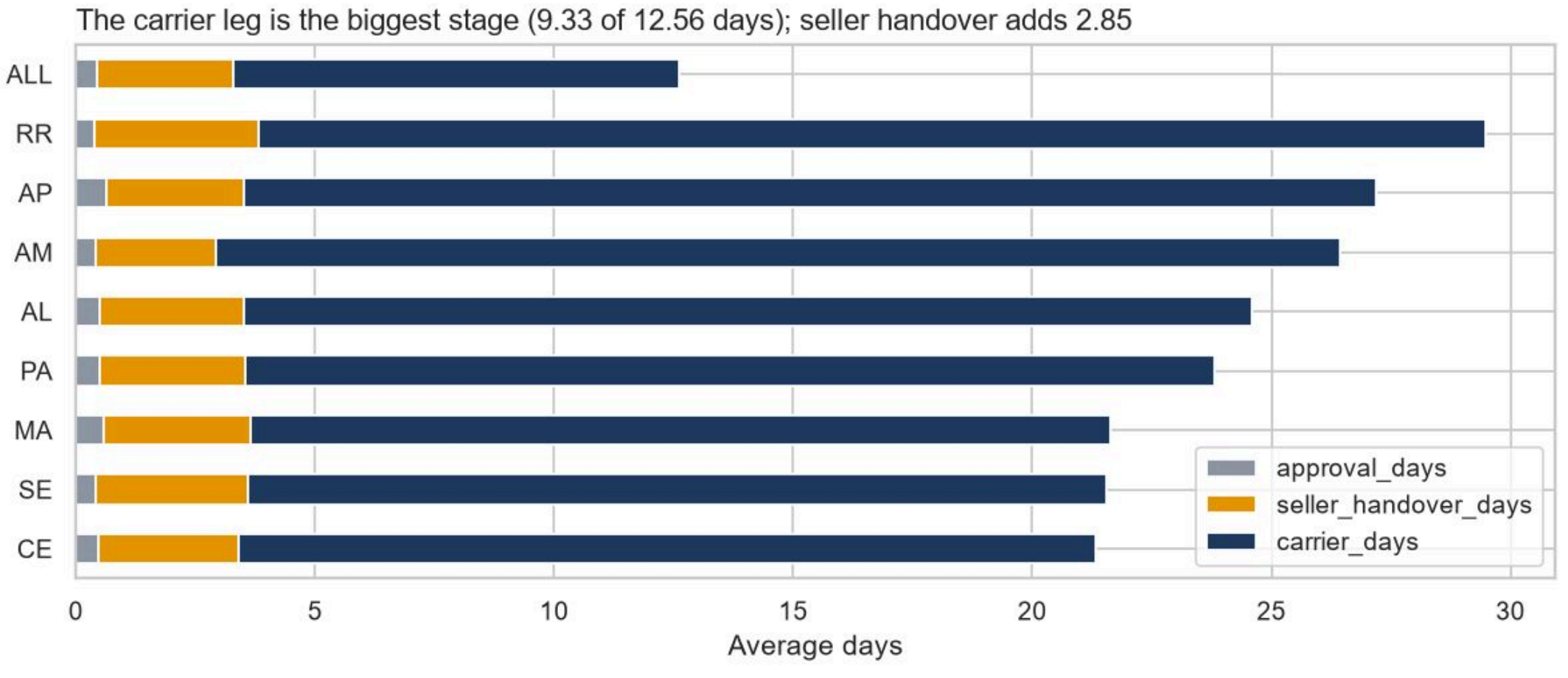
Of the average **12.6-day** delivery window, the carrier leg accounts for **9.3 days** — nearly three-quarters of total time.

Inter-State vs Same-State

15.1 days inter-state vs **7.9 days** same-state — a **+6.9 day** penalty in the regression model.

Handover Multiplier

Each extra seller handover day adds approximately **+1 delivery day** to the customer's wait.



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Who: A Concentrated Group of Sellers

110

Sellers on the Watch List

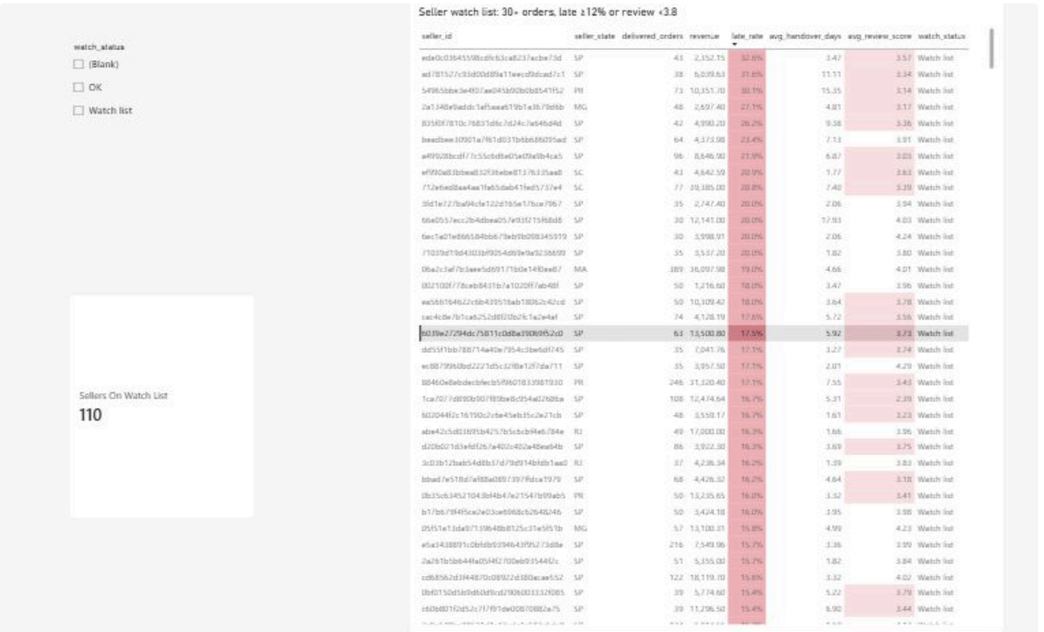
Sellers with 30+ orders who have a late rate ≥ 12% *or* a review score below 3.8.

Their late rate: **12.2%**
Everyone else: **6.1%**

That is **2× the platform average** — concentrated in a small, identifiable cohort.

Why This Is Actionable

The watch list is small enough to manage individually but large enough to move the overall KPI meaningfully.



1 Identify

110 sellers already flagged by the MySQL scorecard query.

2 Engage

Structured outreach: share their scorecard, delete “90-day”, or keep it only if you’d defend it as your own proposal.

3 Monitor

Power BI seller scorecard with drill-through — track progress weekly.

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Recommendations, Sized

Four targeted actions, each with a projected impact on on-time rate and negative review volume.

1

Peak-Season Capacity Plan

KPI: On-time % — lifts on-time from 93.2% to **95.6%**, avoiding an estimated **1,200 negative reviews**.

2

Seller Handover Rule (Max 3 Days)

KPI: On-time % — lifts on-time to **94.6%**, avoiding an estimated **680 negative reviews**.

3

Watch-List Programme (110 Sellers)

KPI: Late %, negative reviews — lifts on-time to **93.9%**, avoiding an estimated **350 negative reviews**.

4

Route-Based Delivery Estimates

Pilot only — a blanket 3-day cut would drop on-time to **88.8%**. Requires route-level calibration before rollout.

Proposed target: on-time at least **95%** every month.

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Dashboard Demo

A fully integrated analytics environment — from executive summary to individual seller drill-through.

Executive Summary
Power BI top-line KPIs

Seller Scorecard
Individual seller metrics

Delivery View
Region slicer filters results

Tableau Map
Click filter and drill-through

Power BI Executive

Top-line KPIs: on-time rate, average review, revenue. Refreshes on data load.

Delivery View

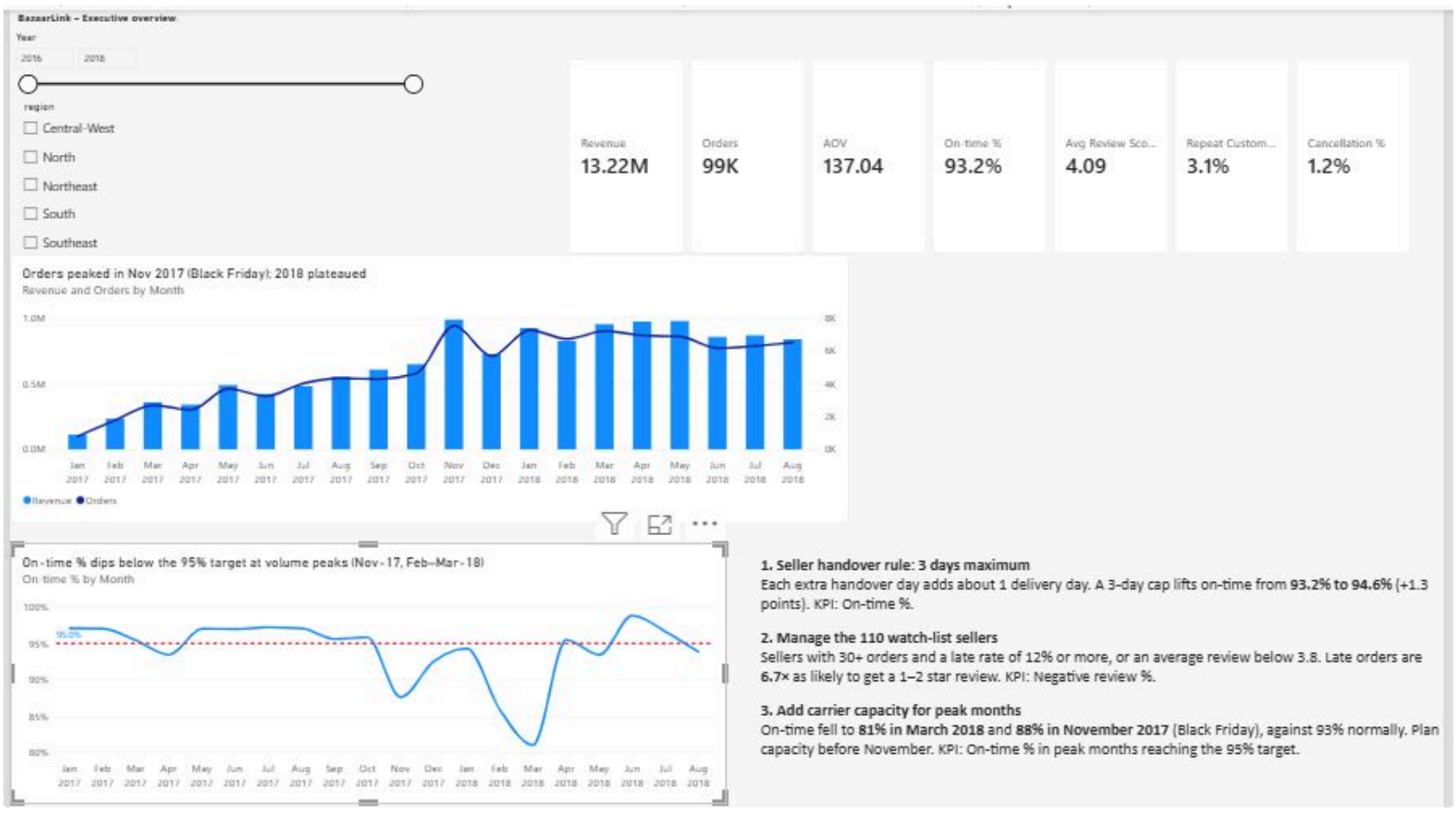
Region slicer filters all visuals simultaneously. Spot state-level performance gaps instantly.

Seller Scorecard

Right-click drill-through to individual seller detail. Watch-list sellers flagged automatically.

Tableau Map

Click any region on the map to filter the full report. Geographic delivery time patterns at a glance.



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Next Steps & Limits

Proposed Actions

→ Approve Actions 1–3

Peak-season plan, handover rule, and watch-list programme are ready for sign-off.

→ Pilot Action 4

Route-based delivery estimates require a controlled pilot before platform-wide rollout.

→ Roll Out KPIs

Unified KPI definitions across all teams — supported by a change management plan.

Tableau dashboard:
public.tableau.com/app/profile/nepstina.s/viz/bazaarlink_tableau_public/Dashboard
Confluence: **nepstina1323-1793988703125.atlassian.net/wiki/spaces/BA**
GitHub:

Known Limits

Associations, Not Causes

The analysis identifies strong correlations. Causal claims require controlled experimentation.

Missing Variables

No distance or carrier-level data available. These would improve model precision.

Model Fit: $R^2 = 0.28$

The regression explains 28% of delivery time variance. Other factors remain unaccounted for.

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